

# $U(1)$ Gauge Field Around a Black Hole

## 1 Quantum Corrections to Maxwell's Equations

The type of corrections to the maxwell equations we will investigate is

$$\nabla_\mu F^{\mu\nu} + \xi^2 R^{\mu\nu}{}_{\sigma\tau} \nabla_\mu F^{\sigma\tau} = 0 \quad (1)$$

In Schwarzschild space the curvature tensor is

$$R^{\mu\nu}{}_{\sigma\tau} = \frac{M}{r^3} (1 - 3(\delta_0^\mu \delta_1^\nu + \delta_1^\mu \delta_0^\nu + \delta_2^\mu \delta_3^\nu + \delta_3^\mu \delta_2^\nu)) (\delta_\sigma^\mu \delta_\tau^\nu - \delta_\tau^\mu \delta_\sigma^\nu) \quad (2)$$

$$\begin{aligned} \nabla_\mu F^{\sigma\tau} &= \nabla_\mu (g^{\sigma\lambda} g^{\tau\rho} (\nabla_\lambda A_\rho - \nabla_\rho A_\lambda)) = g^{\sigma\lambda} g^{\tau\rho} [\nabla_\mu \nabla_\lambda A_\rho - \nabla_\mu \nabla_\rho A_\lambda] \\ &= g^{\sigma\lambda} g^{\tau\rho} [\partial_\mu \nabla_\lambda A_\rho - \Gamma_{\lambda\mu}^\omega \nabla_\omega A_\rho - \Gamma_{\rho\mu}^\omega \nabla_\omega A_\lambda - \partial_\mu \nabla_\rho A_\lambda + \Gamma_{\rho\mu}^\omega \nabla_\omega A_\lambda + \Gamma_{\lambda\mu}^\omega \nabla_\rho A_\omega] \\ &= g^{\sigma\lambda} g^{\tau\rho} [\partial_\mu (\partial_\lambda A_\rho - \Gamma_{\rho\lambda}^\sigma A_\sigma) - \Gamma_{\lambda\mu}^\omega (\partial_\omega A_\rho - \Gamma_{\rho\omega}^\sigma A_\sigma) - \Gamma_{\rho\mu}^\omega (\partial_\lambda A_\omega - \Gamma_{\omega\lambda}^\sigma A_\sigma) \\ &\quad - \partial_\mu (\partial_\rho A_\lambda - \Gamma_{\lambda\rho}^\sigma A_\sigma) + \Gamma_{\rho\mu}^\omega (\partial_\omega A_\lambda - \Gamma_{\lambda\omega}^\sigma A_\sigma) + \Gamma_{\lambda\mu}^\omega (\partial_\rho A_\omega - \Gamma_{\omega\rho}^\sigma A_\sigma)] \\ &= g^{\sigma\lambda} g^{\tau\rho} [\partial_\mu \partial_\lambda A_\rho - \Gamma_{\lambda\mu}^\omega \partial_\omega A_\rho - \Gamma_{\rho\mu}^\omega \partial_\lambda A_\omega - \partial_\mu \partial_\rho A_\lambda + \Gamma_{\rho\mu}^\omega \partial_\omega A_\lambda + \Gamma_{\lambda\mu}^\omega \partial_\rho A_\omega] \\ &= g^{\sigma\lambda} g^{\tau\rho} [\partial_\mu (\partial_\lambda A_\rho - \partial_\rho A_\lambda) - \Gamma_{\lambda\mu}^\omega (\partial_\omega A_\rho - \partial_\rho A_\omega) - \Gamma_{\rho\mu}^\omega (\partial_\lambda A_\omega - \partial_\omega A_\lambda)] \end{aligned} \quad (3)$$

$$\begin{aligned} &(\delta_\sigma^\mu \delta_\tau^\nu - \delta_\tau^\mu \delta_\sigma^\nu) \nabla_\mu F^{\sigma\tau} \\ &= g^{\mu\lambda} g^{\nu\rho} [\partial_\mu (\partial_\lambda A_\rho - \partial_\rho A_\lambda) - \Gamma_{\lambda\mu}^\omega (\partial_\omega A_\rho - \partial_\rho A_\omega) - \Gamma_{\rho\mu}^\omega (\partial_\lambda A_\omega - \partial_\omega A_\lambda)] \\ &\quad - g^{\nu\lambda} g^{\mu\rho} [\partial_\mu (\partial_\lambda A_\rho - \partial_\rho A_\lambda) - \Gamma_{\lambda\mu}^\omega (\partial_\omega A_\rho - \partial_\rho A_\omega) - \Gamma_{\rho\mu}^\omega (\partial_\lambda A_\omega - \partial_\omega A_\lambda)] \\ &= g^{\mu\lambda} g^{\nu\rho} [\partial_\mu (\partial_\lambda A_\rho - \partial_\rho A_\lambda) - \Gamma_{\lambda\mu}^\omega (\partial_\omega A_\rho - \partial_\rho A_\omega) - \Gamma_{\rho\mu}^\omega (\partial_\lambda A_\omega - \partial_\omega A_\lambda)] \\ &\quad - g^{\nu\rho} g^{\mu\lambda} [\partial_\mu (\partial_\rho A_\lambda - \partial_\lambda A_\rho) - \Gamma_{\rho\mu}^\omega (\partial_\omega A_\lambda - \partial_\lambda A_\omega) - \Gamma_{\lambda\mu}^\omega (\partial_\rho A_\omega - \partial_\omega A_\rho)] \\ &= g^{\mu\lambda} g^{\nu\rho} [\partial_\mu (\partial_\lambda A_\rho - \partial_\rho A_\lambda) - \Gamma_{\lambda\mu}^\omega (\partial_\omega A_\rho - \partial_\rho A_\omega) - \Gamma_{\rho\mu}^\omega (\partial_\lambda A_\omega - \partial_\omega A_\lambda)] \\ &\quad + g^{\nu\rho} g^{\mu\lambda} [\partial_\mu (\partial_\lambda A_\rho - \partial_\rho A_\lambda) - \Gamma_{\rho\mu}^\omega (\partial_\lambda A_\omega - \partial_\omega A_\lambda) - \Gamma_{\lambda\mu}^\omega (\partial_\omega A_\rho - \partial_\rho A_\omega)] \\ &= 2g^{\mu\lambda} g^{\nu\rho} [\partial_\mu (\partial_\lambda A_\rho - \partial_\rho A_\lambda) - \Gamma_{\lambda\mu}^\omega (\partial_\omega A_\rho - \partial_\rho A_\omega) - \Gamma_{\rho\mu}^\omega (\partial_\lambda A_\omega - \partial_\omega A_\lambda)] \\ &= 2g^{\mu\lambda} g^{\nu\rho} [LA_\rho - \partial_\mu \partial_\rho A_\lambda - \Gamma_{\lambda\mu}^\omega (-\partial_\rho A_\omega) - \Gamma_{\rho\mu}^\omega (\partial_\lambda A_\omega - \partial_\omega A_\lambda)] \end{aligned} \quad (4)$$

This is just

$$(\delta_\sigma^\mu \delta_\tau^\nu - \delta_\tau^\mu \delta_\sigma^\nu) \nabla_\mu F^{\sigma\tau} = \nabla_\mu F^{\mu\nu} - \nabla_\mu F^{\nu\mu} = 2\nabla_\mu F^{\mu\nu} \quad (5)$$

$$\begin{aligned}
& (1 + \frac{2\xi^2 M}{r^3} (1 - 3(\delta_0^\mu \delta_1^\nu + \delta_1^\mu \delta_0^\nu + \delta_2^\mu \delta_3^\nu + \delta_3^\mu \delta_2^\nu))) \nabla_\mu F^{\mu\nu} \\
&= (1 + \frac{2\xi^2 M}{r^3}) \nabla_\mu F^{\mu\nu} - \frac{6\xi^2 M}{r^3} (\delta_0^\mu \delta_1^\nu + \delta_1^\mu \delta_0^\nu + \delta_2^\mu \delta_3^\nu + \delta_3^\mu \delta_2^\nu) \nabla_\mu F^{\mu\nu} \\
&= (1 + \frac{2\xi^2 M}{r^3}) \nabla_\mu F^{\mu\nu} - \frac{6\xi^2 M}{r^3} (\delta_1^\nu \nabla_0 F^{01} + \delta_0^\nu \nabla_1 F^{10} + \delta_3^\nu \nabla_2 F^{23} + \delta_2^\nu \nabla_3 F^{32}) \\
&= (1 + \frac{2\xi^2 M}{r^3}) \nabla_\mu F^{\mu\nu} + \frac{6\xi^2 M}{r^3} (\delta_0^\nu \nabla_1 F^{01} - \delta_1^\nu \nabla_0 F^{01} + \delta_2^\nu \nabla_3 F^{23} - \delta_3^\nu \nabla_2 F^{23}) = 0 \quad (6)
\end{aligned}$$

The covariant version is

$$(1 + \frac{2\xi^2 M}{r^3}) g^{\lambda\mu} \nabla_\lambda F_{\mu\nu} + \frac{6\xi^2 M}{r^3} g_{\nu\lambda} (\delta_0^\lambda \nabla_1 F^{01} - \delta_1^\lambda \nabla_0 F^{01} + \delta_2^\lambda \nabla_3 F^{23} - \delta_3^\lambda \nabla_2 F^{23}) = 0 \quad (7)$$

The quantity  $\nabla_1 F^{01}$  is

$$\begin{aligned}
\nabla_1 F^{01} &= g^{0\lambda} g^{1\rho} [\partial_1 (\partial_\lambda A_\rho - \partial_\rho A_\lambda) - \Gamma_{\lambda 1}^\omega (\partial_\omega A_\rho - \partial_\rho A_\omega) - \Gamma_{\rho 1}^\omega (\partial_\lambda A_\omega - \partial_\omega A_\lambda)] \\
&= g^{00} g^{11} [\partial_1 (\partial_0 A_1 - \partial_1 A_0) - \Gamma_{01}^\omega (\partial_\omega A_1 - \partial_1 A_\omega) - \Gamma_{11}^\omega (\partial_0 A_\omega - \partial_\omega A_0)] \quad (8)
\end{aligned}$$

Note that

$$\Gamma_{01}^\omega = \left\{ \frac{M}{r^2(1 - \frac{2M}{r})}, 0, 0, 0 \right\}, \quad \Gamma_{11}^\omega = \left\{ 0, -\frac{M}{r^2(1 - \frac{2M}{r})}, 0, 0 \right\} \quad (9)$$

Thus

$$\begin{aligned}
\nabla_1 F^{01} &= g^{00} g^{11} \left[ \partial_1 (\partial_0 A_1 - \partial_1 A_0) - \frac{M}{r^2(1 - \frac{2M}{r})} (\partial_0 A_1 - \partial_1 A_0) + \frac{M}{r^2(1 - \frac{2M}{r})} (\partial_0 A_1 - \partial_1 A_0) \right] \\
&= g^{00} g^{11} (\partial_1 \partial_0 A_1 - \partial_1^2 A_0) = \partial_r^2 A_t - \partial_r \partial_t A_r \quad (10)
\end{aligned}$$

The quantity  $\nabla_0 F^{01}$  is

$$\begin{aligned}
\nabla_0 F^{01} &= g^{0\lambda} g^{1\rho} [\partial_0 (\partial_\lambda A_\rho - \partial_\rho A_\lambda) - \Gamma_{\lambda 0}^\omega (\partial_\omega A_\rho - \partial_\rho A_\omega) - \Gamma_{\rho 0}^\omega (\partial_\lambda A_\omega - \partial_\omega A_\lambda)] \\
&= g^{00} g^{11} [\partial_0 (\partial_0 A_1 - \partial_1 A_0) - \Gamma_{00}^\omega (\partial_\omega A_1 - \partial_1 A_\omega) - \Gamma_{10}^\omega (\partial_0 A_\omega - \partial_\omega A_0)] \quad (11)
\end{aligned}$$

$$\Gamma_{00}^\omega = \left\{ 0, \frac{M}{r^2} (1 - \frac{2M}{r}), 0, 0 \right\} \quad (12)$$

Thus

$$\begin{aligned}
\nabla_0 F^{01} &= g^{00} g^{11} [\partial_0 (\partial_0 A_1 - \partial_1 A_0) - \Gamma_{00}^1 (\partial_1 A_1 - \partial_1 A_1) - \Gamma_{10}^0 (\partial_0 A_0 - \partial_0 A_0)] \\
&= -\partial_0 (\partial_0 A_1 - \partial_1 A_0) = \partial_t \partial_r A_t - \partial_t^2 A_r \quad (13)
\end{aligned}$$

The quantity  $\nabla_3 F^{23}$  is

$$\begin{aligned}\nabla_3 F^{23} &= g^{2\lambda} g^{3\rho} [\partial_3(\partial_\lambda A_\rho - \partial_\rho A_\lambda) - \Gamma_{\lambda 3}^\omega(\partial_\omega A_\rho - \partial_\rho A_\omega) - \Gamma_{\rho 3}^\omega(\partial_\lambda A_\omega - \partial_\omega A_\lambda)] \\ &= g^{23} g^{32} [\partial_3(\partial_3 A_2 - \partial_2 A_3) - \Gamma_{33}^\omega(\partial_\omega A_2 - \partial_2 A_\omega) - \Gamma_{23}^\omega(\partial_3 A_\omega - \partial_\omega A_3)]\end{aligned}\quad (14)$$

Note that

$$\Gamma_{23}^\omega = \left\{ 0, -\frac{2r(1 - \frac{2M}{r})}{(1 + z\bar{z})^2}, 0, 0 \right\}, \quad \Gamma_{33}^\omega = \left\{ 0, 0, 0, -\frac{2z}{1 + z\bar{z}} \right\} \quad (15)$$

Thus

$$\begin{aligned}\nabla_3 F^{23} &= \frac{(1 + z\bar{z})^4}{4r^4} \left[ \partial_3^2 A_2 - \partial_3 \partial_2 A_3 + \frac{2z}{1 + z\bar{z}}(\partial_3 A_2 - \partial_2 A_3) + \frac{2r(1 - \frac{2M}{r})}{(1 + z\bar{z})^2}(\partial_3 A_1 - \partial_1 A_3) \right] \\ &= \frac{(1 + z\bar{z})^4}{4r^4} \left[ \partial_{\bar{z}}^2 A_z - \partial_{\bar{z}} \partial_z A_{\bar{z}} + \frac{2z}{1 + z\bar{z}}(\partial_{\bar{z}} A_z - \partial_z A_{\bar{z}}) + \frac{2r(1 - \frac{2M}{r})}{(1 + z\bar{z})^2}(\partial_{\bar{z}} A_r - \partial_r A_{\bar{z}}) \right]\end{aligned}\quad (16)$$

The quantity  $\nabla_2 F^{23}$  is

$$\begin{aligned}\nabla_2 F^{23} &= g^{2\lambda} g^{3\rho} [\partial_2(\partial_\lambda A_\rho - \partial_\rho A_\lambda) - \Gamma_{\lambda 2}^\omega(\partial_\omega A_\rho - \partial_\rho A_\omega) - \Gamma_{\rho 2}^\omega(\partial_\lambda A_\omega - \partial_\omega A_\lambda)] \\ &= g^{23} g^{32} [\partial_2(\partial_3 A_2 - \partial_2 A_3) - \Gamma_{32}^\omega(\partial_\omega A_2 - \partial_2 A_\omega) - \Gamma_{22}^\omega(\partial_3 A_\omega - \partial_\omega A_3)]\end{aligned}\quad (17)$$

Note that

$$\Gamma_{22}^\omega = \left\{ 0, 0, -\frac{2\bar{z}}{1 + z\bar{z}}, 0 \right\} \quad (18)$$

Thus

$$\begin{aligned}\nabla_2 F^{23} &= \frac{(1 + z\bar{z})^4}{4r^4} \left[ \partial_2 \partial_3 A_2 - \partial_2^2 A_3 + \frac{2r(1 - \frac{2M}{r})}{(1 + z\bar{z})^2}(\partial_1 A_2 - \partial_2 A_1) + \frac{2\bar{z}}{1 + z\bar{z}}(\partial_3 A_2 - \partial_2 A_3) \right] \\ &= -\frac{(1 + z\bar{z})^4}{4r^4} \left[ \partial_z^2 A_{\bar{z}} - \partial_z \partial_{\bar{z}} A_z + \frac{2\bar{z}}{1 + z\bar{z}}(\partial_z A_{\bar{z}} - \partial_{\bar{z}} A_z) + \frac{2r(1 - \frac{2M}{r})}{(1 + z\bar{z})^2}(\partial_z A_r - \partial_r A_z) \right]\end{aligned}\quad (19)$$

## 1.1 Equations of Motion

The equation of motion for the  $\nu = 0$  component of the covariant vector potential is

$$\begin{aligned}(1 + \frac{2\xi^2 M}{r^3})g^{\lambda\mu}\nabla_\lambda F_{\mu 0} + \frac{6\xi^2 M}{r^3}g_{0\lambda}\delta_0^\lambda\nabla_1 F^{01} &= 0 \\ \Rightarrow LA_t - \frac{2M}{r^2}(\partial_t A_r - \partial_r A_t) - \partial_t \nabla_\mu A^\mu &= -\frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}(1 - \frac{2M}{r})(\partial_r^2 A_t - \partial_r \partial_t A_r)\end{aligned}\quad (20)$$

For the  $\nu = 1$  component the equation of motion is

$$\begin{aligned}
& (1 + \frac{2\xi^2 M}{r^3})g^{\lambda\mu}\nabla_\lambda F_{\mu 1} - \frac{6\xi^2 M}{r^3}g_{1\lambda}\delta_1^\lambda\nabla_0 F^{01} = 0 \\
\Rightarrow & LA_r - \frac{2M}{r^2(1 - \frac{2M}{r})^2}\partial_t A_t - \frac{2M}{r^2}\partial_r A_r + \frac{(1 + z\bar{z})^2}{r^3}(\partial_z A_{\bar{z}} + \partial_{\bar{z}} A_z) \\
& + \frac{2}{r^2}(1 - \frac{2M}{r})A_r - \partial_r \nabla_\mu A^\mu = \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}(1 - \frac{2M}{r})^{-1}(\partial_t^2 A_r - \partial_t \partial_r A_t)
\end{aligned} \tag{21}$$

For the  $\nu = 2$  component the equation of motion is

$$\begin{aligned}
& (1 + \frac{2\xi^2 M}{r^3})g^{\lambda\mu}\nabla_\lambda F_{\mu 2} - \frac{6\xi^2 M}{r^3}g_{2\lambda}\delta_3^\lambda\nabla_2 F^{23} = 0 \\
\Rightarrow & LA_{\bar{z}} + \frac{2}{r}(1 - \frac{2M}{r})(\partial_r A_{\bar{z}} - \partial_z A_r) - \frac{2\bar{z}(1 + z\bar{z})}{r^2}\partial_{\bar{z}} A_z - \partial_z \nabla_\mu A^\mu \\
= & \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}\frac{(1 + z\bar{z})^2}{2r^2}\left[\partial_z^2 A_{\bar{z}} - \partial_z \partial_{\bar{z}} A_z + \frac{2\bar{z}}{1 + z\bar{z}}(\partial_z A_{\bar{z}} - \partial_{\bar{z}} A_z) + \frac{2r(1 - \frac{2M}{r})}{(1 + z\bar{z})^2}(\partial_z A_r - \partial_r A_z)\right]
\end{aligned} \tag{22}$$

And finally, for the  $\nu = 3$  component the equation of motion is

$$\begin{aligned}
& (1 + \frac{2\xi^2 M}{r^3})g^{\lambda\mu}\nabla_\lambda F_{\mu 3} + \frac{6\xi^2 M}{r^3}g_{3\lambda}\delta_2^\lambda\nabla_3 F^{23} = 0 \\
\Rightarrow & LA_z + \frac{2}{r}(1 - \frac{2M}{r})(\partial_r A_{\bar{z}} - \partial_{\bar{z}} A_r) - \frac{2z(1 + z\bar{z})}{r^2}\partial_z A_{\bar{z}} - \partial_{\bar{z}} \nabla_\mu A^\mu \\
= & \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}\frac{(1 + z\bar{z})^2}{2r^2}\left[\partial_{\bar{z}}^2 A_z - \partial_{\bar{z}} \partial_z A_{\bar{z}} + \frac{2z}{1 + z\bar{z}}(\partial_{\bar{z}} A_z - \partial_z A_{\bar{z}}) + \frac{2r(1 - \frac{2M}{r})}{(1 + z\bar{z})^2}(\partial_{\bar{z}} A_r - \partial_r A_{\bar{z}})\right]
\end{aligned} \tag{23}$$

## 1.2 Gauge Fixing

Let us choose the gauge  $A_t = 0$ . The quantity  $\nabla_\mu A^\mu$  becomes

$$\nabla_\mu A^\mu = -\frac{1}{r^2}(1 - \frac{2M}{r})\partial_r(r^2 A_r) - \frac{2M}{r^2}A_r - \frac{(1 + z\bar{z})^2}{2r^2}(\partial_z A_{\bar{z}} + \partial_{\bar{z}} A_z) \tag{24}$$

The constraint becomes

$$\begin{aligned}
& -\frac{2M}{r^2}\partial_t A_r - \partial_t \nabla_\mu A^\mu = \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}(1 - \frac{2M}{r})\partial_r \partial_t A_r \\
\Rightarrow & \partial_t \left[ \nabla_\mu A^\mu + \frac{2M}{r^2}A_r + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}(1 - \frac{2M}{r})\partial_r A_r \right] = 0 \\
\Rightarrow & \nabla_\mu A^\mu + \frac{2M}{r^2}A_r + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}(1 - \frac{2M}{r})\partial_r A_r = 0
\end{aligned} \tag{25}$$

Combining

$$\begin{aligned}
& \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \left(1 - \frac{2M}{r}\right) \partial_r A_r = \frac{1}{r^2} \left(1 - \frac{2M}{r}\right) \partial_r (r^2 A_r) + \frac{(1 + z\bar{z})^2}{2r^2} (\partial_z A_{\bar{z}} + \partial_{\bar{z}} A_z) \\
\Rightarrow & -r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right) \partial_r A_r = 2r \left(1 + \frac{2\xi^2 M}{r^3}\right) A_r + \frac{(1 + z\bar{z})^2}{2\left(1 - \frac{2M}{r}\right)} \left(1 + \frac{2\xi^2 M}{r^3}\right) (\partial_z A_{\bar{z}} + \partial_{\bar{z}} A_z) \\
\Rightarrow & -r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right) \partial_r A_r = \partial_r \left[ r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right) \right] A_r + \frac{(1 + z\bar{z})^2}{2\left(1 - \frac{2M}{r}\right)} \left(1 + \frac{2\xi^2 M}{r^3}\right) (\partial_z A_{\bar{z}} + \partial_{\bar{z}} A_z) \\
\Rightarrow & \partial_r \left[ r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right) A_r \right] + \frac{(1 + z\bar{z})^2}{2\left(1 - \frac{2M}{r}\right)} \left(1 + \frac{2\xi^2 M}{r^3}\right) (\partial_z A_{\bar{z}} + \partial_{\bar{z}} A_z) = 0 \\
\Rightarrow & \frac{1}{\left(1 + \frac{2\xi^2 M}{r^3}\right)} \partial_r \left[ r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right) A_r \right] = -\frac{(1 + z\bar{z})^2}{2\left(1 - \frac{2M}{r}\right)} (\partial_z A_{\bar{z}} + \partial_{\bar{z}} A_z) \quad (26)
\end{aligned}$$

Let's look at the  $A_r$  equation of motion

$$\begin{aligned}
& LA_r - \frac{2M}{r^2} \partial_r A_r + \frac{(1+z\bar{z})^2}{r^3} (\partial_z A_{\bar{z}} + \partial_{\bar{z}} A_z) \\
& + \frac{2}{r^2} (1 - \frac{2M}{r}) A_r - \partial_r \nabla_\mu A^\mu = \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} (1 - \frac{2M}{r})^{-1} \partial_t^2 A_r \\
\Rightarrow & LA_r - \frac{2M}{r^2} \partial_r A_r - \frac{2(1 - \frac{2M}{r})}{r^3(1 + \frac{2\xi^2 M}{r^3})} \partial_r \left[ r^2 (1 - \frac{4\xi^2 M}{r^3}) A_r \right] \\
& + \frac{2}{r^2} (1 - \frac{2M}{r}) A_r + \partial_r \left( \frac{2M}{r^2} A_r + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} (1 - \frac{2M}{r}) \partial_r A_r \right) = \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} (1 - \frac{2M}{r})^{-1} \partial_t^2 A_r \\
\Rightarrow & LA_r - \frac{2}{r^3} (1 - \frac{2M}{r}) \partial_r (r^2 A_r) + \frac{2}{r^2} (1 - \frac{2M}{r}) A_r + \partial_r \left( \frac{2M}{r^2} A_r \right) \\
& + \frac{2}{r^3} (1 - \frac{2M}{r}) \partial_r (r^2 A_r) - \frac{2(1 - \frac{2M}{r})}{r^3(1 + \frac{2\xi^2 M}{r^3})} \partial_r \left[ r^2 (1 + \frac{2\xi^2 M}{r^3}) A_r \right] + \frac{2(1 - \frac{2M}{r})}{r^3(1 + \frac{2\xi^2 M}{r^3})} \partial_r \left[ r^2 \frac{6\xi^2 M}{r^3} A_r \right] \\
& + \partial_r \left( \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} (1 - \frac{2M}{r}) \partial_r A_r \right) = \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} (1 - \frac{2M}{r})^{-1} \partial_t^2 A_r \\
\Rightarrow & LA_r - \frac{2}{r^2} \partial_r \left[ r (1 - \frac{2M}{r}) A_r \right] + \frac{2(1 - \frac{2M}{r})}{r^3(1 + \frac{2\xi^2 M}{r^3})} \frac{2\xi^2 M}{r^3} \partial_r (r^2 A_r) + \frac{2(1 - \frac{2M}{r})}{r^3(1 + \frac{2\xi^2 M}{r^3})} \partial_r \left[ r^2 \frac{4\xi^2 M}{r^3} A_r \right] \\
& + \partial_r \left( \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} (1 - \frac{2M}{r}) \partial_r A_r \right) = \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} (1 - \frac{2M}{r})^{-1} \partial_t^2 A_r \\
\Rightarrow & LA_r - \frac{2}{r^2} \partial_r \left[ r (1 - \frac{2M}{r}) A_r \right] \\
& + \frac{2(1 - \frac{2M}{r})}{r^3(1 + \frac{2\xi^2 M}{r^3})} \left[ r^2 \frac{2\xi^2 M}{r^3} \partial_r A_r + 2r \frac{2\xi^2 M}{r^3} A_r + r^2 \frac{4\xi^2 M}{r^3} \partial_r A_r - \frac{4\xi^2 M}{r^2} A_r \right] \\
& + \partial_r \left( \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} (1 - \frac{2M}{r}) \partial_r A_r \right) = \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} (1 - \frac{2M}{r})^{-1} \partial_t^2 A_r \\
\Rightarrow & LA_r - \frac{2}{r^2} \partial_r \left[ r (1 - \frac{2M}{r}) A_r \right] + \frac{2}{r} \frac{\frac{6\xi^2 M}{r^3} (1 - \frac{2M}{r})}{(1 + \frac{2\xi^2 M}{r^3})} \partial_r A_r \\
& + \partial_r \left( \frac{\frac{6\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r A_r \right) = \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} (1 - \frac{2M}{r})^{-1} \partial_t^2 A_r \\
\Rightarrow & LA_r - \frac{2}{r^2} \partial_r \left[ r (1 - \frac{2M}{r}) A_r \right] \\
& + \frac{1}{r^2} \partial_r \left( r^2 \frac{\frac{6\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r A_r \right) = \frac{\frac{6\xi^2 M}{r^3} (1 - \frac{2M}{r})^{-1}}{1 + \frac{2\xi^2 M}{r^3}} \partial_t^2 A_r \tag{27}
\end{aligned}$$

The equation for  $A_z$

$$\begin{aligned}
& LA_z + \frac{2}{r}(1 - \frac{2M}{r})(\partial_r A_z - \partial_z A_r) - \frac{2\bar{z}(1 + z\bar{z})}{r^2} \partial_{\bar{z}} A_z - \partial_z \nabla_\mu A^\mu \\
&= \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{(1 + z\bar{z})^2}{2r^2} \left[ \partial_z^2 A_{\bar{z}} - \partial_z \partial_{\bar{z}} A_z + \frac{2\bar{z}}{1 + z\bar{z}} (\partial_z A_{\bar{z}} - \partial_{\bar{z}} A_z) + \frac{2r(1 - \frac{2M}{r})}{(1 + z\bar{z})^2} (\partial_z A_r - \partial_r A_z) \right] \\
&\Rightarrow \left( L - \frac{2\bar{z}(1 + z\bar{z})}{r^2} \partial_{\bar{z}} \right) A_z = -\frac{2}{r}(1 - \frac{2M}{r})(\partial_r A_z - \partial_z A_r) \\
&\quad - \frac{2M}{r^2} \partial_z A_r - \frac{\frac{6\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_z \partial_r A_r + \frac{1}{r} \frac{\frac{6\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} (\partial_z A_r - \partial_r A_z) \\
&\quad + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{1}{2r^2} [(1 + z\bar{z})^2 \partial_z (\partial_z A_{\bar{z}} - \partial_{\bar{z}} A_z) + 2\bar{z}(1 + z\bar{z})(\partial_z A_{\bar{z}} - \partial_{\bar{z}} A_z)] \\
&\Rightarrow \left( L - \frac{2\bar{z}(1 + z\bar{z})}{r^2} \partial_{\bar{z}} \right) A_z = -\frac{2}{r}(1 - \frac{2M}{r})(\partial_r A_z - \partial_z A_r) \\
&\quad - \frac{2M}{r^2} \partial_z A_r - \frac{\frac{6\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_z \partial_r A_r + \frac{1}{r} \frac{\frac{6\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} (\partial_z A_r - \partial_r A_z) \\
&\quad + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{1}{2r^2} \partial_z [(1 + z\bar{z})^2 (\partial_z A_{\bar{z}} - \partial_{\bar{z}} A_z)] \\
&\Rightarrow \left( L - \frac{2\bar{z}(1 + z\bar{z})}{r^2} \partial_{\bar{z}} \right) A_z = -\frac{2}{r}(1 - \frac{2M}{r}) \left[ 1 + \frac{\frac{3\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \right] \partial_r A_z \\
&\quad + \frac{2}{r} \left[ \left( 1 - \frac{3M}{r} \right) + \frac{\frac{3\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \right] \partial_z A_r - \frac{\frac{6\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_z \partial_r A_r \\
&\quad + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{1}{2r^2} \partial_z [(1 + z\bar{z})^2 (\partial_z A_{\bar{z}} - \partial_{\bar{z}} A_z)] \tag{28}
\end{aligned}$$

The equation for  $A_{\bar{z}}$

$$\begin{aligned}
& LA_{\bar{z}} + \frac{2}{r}(1 - \frac{2M}{r})(\partial_r A_{\bar{z}} - \partial_{\bar{z}} A_r) - \frac{2z(1 + z\bar{z})}{r^2} \partial_z A_{\bar{z}} - \partial_{\bar{z}} \nabla_\mu A^\mu \\
&= \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{(1 + z\bar{z})^2}{2r^2} \left[ \partial_{\bar{z}}^2 A_z - \partial_{\bar{z}} \partial_z A_{\bar{z}} + \frac{2z}{1 + z\bar{z}} (\partial_{\bar{z}} A_z - \partial_z A_{\bar{z}}) + \frac{2r(1 - \frac{2M}{r})}{(1 + z\bar{z})^2} (\partial_{\bar{z}} A_r - \partial_r A_{\bar{z}}) \right] \\
&\Rightarrow \left( L - \frac{2z(1 + z\bar{z})}{r^2} \partial_z \right) A_{\bar{z}} = -\frac{2}{r}(1 - \frac{2M}{r})(\partial_r A_{\bar{z}} - \partial_{\bar{z}} A_r) \\
&\quad - \frac{2M}{r^2} \partial_{\bar{z}} A_r - \frac{\frac{6\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_{\bar{z}} \partial_r A_r + \frac{1}{r} \frac{\frac{6\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} (\partial_{\bar{z}} A_r - \partial_r A_{\bar{z}}) \\
&\quad + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{1}{2r^2} [(1 + z\bar{z})^2 \partial_{\bar{z}} (\partial_{\bar{z}} A_z - \partial_z A_{\bar{z}}) + 2z(1 + z\bar{z})(\partial_{\bar{z}} A_z - \partial_z A_{\bar{z}})] \\
&\Rightarrow \left( L - \frac{2z(1 + z\bar{z})}{r^2} \partial_z \right) A_{\bar{z}} = -\frac{2}{r}(1 - \frac{2M}{r}) \left( 1 + \frac{\frac{3\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \right) \partial_r A_{\bar{z}} \\
&\quad + \frac{2}{r} \left( \left( 1 - \frac{3M}{r} \right) + \frac{\frac{3\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \right) \partial_{\bar{z}} A_r - \frac{\frac{6\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_{\bar{z}} \partial_r A_r \\
&\quad + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{1}{2r^2} \partial_{\bar{z}} [(1 + z\bar{z})^2 (\partial_{\bar{z}} A_z - \partial_z A_{\bar{z}})] \tag{29}
\end{aligned}$$

The gauge fixed equations of motion are

$$\begin{aligned}
& LA_r - \frac{2}{r^2} \partial_r \left[ r(1 - \frac{2M}{r}) A_r \right] \\
&+ \frac{1}{r^2} \partial_r \left( r^2 \frac{\frac{6\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r A_r \right) = \frac{\frac{6\xi^2 M}{r^3}(1 - \frac{2M}{r})^{-1}}{1 + \frac{2\xi^2 M}{r^3}} \partial_t^2 A_r \tag{30}
\end{aligned}$$

$$\begin{aligned}
& \left( L - \frac{2\bar{z}(1 + z\bar{z})}{r^2} \partial_{\bar{z}} \right) A_z = -\frac{2}{r}(1 - \frac{2M}{r}) \left[ 1 + \frac{\frac{3\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \right] \partial_r A_z \\
&+ \frac{2}{r} \left[ \left( 1 - \frac{3M}{r} \right) + \frac{\frac{3\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \right] \partial_z A_r - 2 \frac{\frac{3\xi^2 M}{r^3}(1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_z \partial_r A_r \\
&\quad + \frac{1}{r^2} \frac{\frac{3\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \partial_z [(1 + z\bar{z})^2 (\partial_z A_{\bar{z}} - \partial_{\bar{z}} A_z)] \tag{31}
\end{aligned}$$



$$\begin{aligned}
\left(L - \frac{2z(1+z\bar{z})}{r^2}\partial_z\right)A_{\bar{z}} &= -\frac{2}{r}\left(1 - \frac{2M}{r}\right)\left[1 + \frac{\frac{3\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}\right]\partial_r A_{\bar{z}} \\
&+ \frac{2}{r}\left[\left(1 - \frac{3M}{r}\right) + \frac{\frac{3\xi^2 M}{r^3}\left(1 - \frac{2M}{r}\right)}{1 + \frac{2\xi^2 M}{r^3}}\right]\partial_{\bar{z}}A_r - 2\frac{\frac{3\xi^2 M}{r^3}\left(1 - \frac{2M}{r}\right)}{1 + \frac{2\xi^2 M}{r^3}}\partial_{\bar{z}}\partial_r A_r \\
&+ \frac{1}{r^2}\frac{\frac{3\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}\partial_{\bar{z}}\left[(1+z\bar{z})^2(\partial_{\bar{z}}A_z - \partial_zA_{\bar{z}})\right]
\end{aligned} \tag{32}$$

and the constraint equation is

$$\frac{1}{\left(1 + \frac{2\xi^2 M}{r^3}\right)}\partial_r\left[r^2\left(1 - \frac{4\xi^2 M}{r^3}\right)A_r\right] = -\frac{(1+z\bar{z})^2}{2\left(1 - \frac{2M}{r}\right)}(\partial_zA_{\bar{z}} + \partial_{\bar{z}}A_z) \tag{33}$$

### 1.3 The First Solution

and suppose that  $A_r = 0$ ,  $A_z = \partial_z\Phi$  and  $A_{\bar{z}} = -\partial_{\bar{z}}\Phi$ . Then the  $A_z$  equation becomes

$$\begin{aligned}
&\left(L - \frac{2\bar{z}(1+z\bar{z})}{r^2}\partial_z\right)\partial_z\Phi = -\frac{2}{r}\left(1 - \frac{2M}{r}\right)\partial_r\partial_z\Phi \\
&- \frac{1}{r}\frac{\frac{6\xi^2 M}{r^3}\left(1 - \frac{2M}{r}\right)}{1 + \frac{2\xi^2 M}{r^3}}\partial_r\partial_z\Phi - \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}\frac{1}{r^2}\partial_z\left[(1+z\bar{z})^2\partial_z\partial_{\bar{z}}\Phi\right] \\
\Rightarrow \quad &\partial_z\left[L\Phi + \frac{2}{r}\left(1 - \frac{2M}{r}\right)\partial_r\Phi + \frac{1}{r}\frac{\frac{6\xi^2 M}{r^3}\left(1 - \frac{2M}{r}\right)}{1 + \frac{2\xi^2 M}{r^3}}\partial_r\Phi + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}\frac{(1+z\bar{z})^2}{r^2}\partial_z\partial_{\bar{z}}\Phi\right] = 0 \\
\Rightarrow \quad &L\Phi + \frac{2}{r}\left(1 - \frac{2M}{r}\right)\partial_r\Phi + \frac{1}{r}\frac{\frac{6\xi^2 M}{r^3}\left(1 - \frac{2M}{r}\right)}{1 + \frac{2\xi^2 M}{r^3}}\partial_r\Phi + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}\frac{(1+z\bar{z})^2}{r^2}\partial_z\partial_{\bar{z}}\Phi = 0 \tag{34}
\end{aligned}$$

$$\begin{aligned}
& (1 - \frac{2M}{r})^{-1} \partial_t^2 \Phi - \frac{1}{r^2} \partial_r \left( r^2 (1 - \frac{2M}{r}) \partial_r \Phi \right) - \frac{(1 + z\bar{z})^2}{r^2} \partial_z \partial_{\bar{z}} \Phi \\
& + \frac{2}{r} (1 - \frac{2M}{r}) \partial_r \Phi + \frac{1}{r} \frac{\frac{6\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \Phi + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{(1 + z\bar{z})^2}{r^2} \partial_z \partial_{\bar{z}} \Phi = 0 \\
\Rightarrow & (1 - \frac{2M}{r})^{-1} \partial_t^2 \Phi - \partial_r \left( (1 - \frac{2M}{r}) \partial_r \Phi \right) + \frac{l(l+1)}{r^2} \Phi \\
& + \frac{1}{r} \frac{\frac{6\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \Phi - \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{l(l+1)}{r^2} \Phi = 0 \\
\Rightarrow & \partial_t^2 \Phi - (1 - \frac{2M}{r}) \partial_r \left( (1 - \frac{2M}{r}) \partial_r \Phi \right) \\
& + \frac{2}{r} \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})^2}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \Phi + \frac{l(l+1)}{r^2} (1 - \frac{2M}{r}) \frac{1 - \frac{4\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \Phi = 0 \\
\Rightarrow & \partial_t^2 \Phi - \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \left( (1 - \frac{2M}{r}) (1 + \frac{2\xi^2 M}{r^3}) \partial_r \Phi \right) + \frac{l(l+1)}{r^2} (1 - \frac{2M}{r}) \frac{1 - \frac{4\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \Phi = 0
\end{aligned} \tag{35}$$

We must check that this is consistent with the  $A_{\bar{z}}$  equation:

$$\begin{aligned}
& - \left( L - \frac{2z(1 + z\bar{z})}{r^2} \partial_z \right) \partial_{\bar{z}} \Phi = \frac{2}{r} (1 - \frac{2M}{r}) \partial_r \partial_{\bar{z}} \Phi \\
& + \frac{1}{r} \frac{\frac{6\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \partial_{\bar{z}} \Phi + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{1}{r^2} \partial_{\bar{z}} [(1 + z\bar{z})^2 \partial_z \partial_{\bar{z}} \Phi] \\
\Rightarrow & \partial_{\bar{z}} \left[ L\Phi + \frac{2}{r} (1 - \frac{2M}{r}) \partial_r \Phi + \frac{1}{r} \frac{\frac{6\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \Phi + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{(1 + z\bar{z})^2}{r^2} \partial_z \partial_{\bar{z}} \Phi \right] = 0 \\
\Rightarrow & L\Phi + \frac{2}{r} (1 - \frac{2M}{r}) \partial_r \Phi + \frac{1}{r} \frac{\frac{6\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \Phi + \frac{\frac{6\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \frac{(1 + z\bar{z})^2}{r^2} \partial_z \partial_{\bar{z}} \Phi = 0
\end{aligned} \tag{36}$$

This is consistent with (34)

## 1.4 A Second Solution

This time let's suppose that  $A_z = \partial_z \Psi$  and  $A_{\bar{z}} = \partial_{\bar{z}} \Psi$ . The constraint equation becomes

$$\frac{1}{(1 + \frac{2\xi^2 M}{r^3})} \partial_r \left[ r^2 (1 - \frac{4\xi^2 M}{r^3}) A_r \right] = - \frac{(1 + z\bar{z})^2}{(1 - \frac{2M}{r})} \partial_z \partial_{\bar{z}} \Psi \tag{37}$$

If  $\Psi \propto Y_l^m$  then the constraint becomes

$$\partial_r \left[ r^2 (1 - \frac{4\xi^2 M}{r^3}) A_r \right] = \frac{l(l+1)(1 + \frac{2\xi^2 M}{r^3})}{(1 - \frac{2M}{r})} \Psi \tag{38}$$

This implies that

$$A_r = \frac{l(l+1)\Phi}{r^2(1 - \frac{4\xi^2 M}{r^3})}, \quad \Psi = \frac{(1 - \frac{2M}{r})\partial_r \Phi}{1 + \frac{2\xi^2 M}{r^3}} \quad (39)$$

$$\begin{aligned} & \left( L - \frac{2\bar{z}(1+z\bar{z})}{r^2} \partial_{\bar{z}} \right) \partial_z \Psi = -\frac{2}{r} \left( 1 - \frac{2M}{r} \right) \left[ 1 + \frac{\frac{3\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \right] \partial_r \partial_z \Psi \\ & + \frac{2}{r} \left[ \left( 1 - \frac{3M}{r} \right) + \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \right] \partial_z A_r - 2 \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_z \partial_r A_r \\ & \Rightarrow \quad \partial_z L \Psi = -\frac{2}{r} \left( 1 - \frac{2M}{r} \right) \left[ 1 + \frac{\frac{3\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \right] \partial_r \partial_z \Psi \\ & + \frac{2}{r} \left[ \left( 1 - \frac{3M}{r} \right) + \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \right] \partial_z A_r - 2 \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_z \partial_r A_r \\ & \Rightarrow \quad L \Psi = -\frac{2}{r} \left( 1 - \frac{2M}{r} \right) \left[ 1 + \frac{\frac{3\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \right] \partial_r \Psi \\ & + \frac{2}{r} \left[ \left( 1 - \frac{3M}{r} \right) + \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \right] A_r - 2 \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r A_r \\ \Rightarrow & \quad \left( 1 - \frac{2M}{r} \right)^{-1} \partial_t^2 \Psi - \frac{1}{r^2} \partial_r \left( r^2 \left( 1 - \frac{2M}{r} \right) \partial_r \Psi \right) + \frac{l(l+1)}{r^2} \Psi = -\frac{2}{r} \left( 1 - \frac{2M}{r} \right) \left[ 1 + \frac{\frac{3\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \right] \partial_r \Psi \\ & + \frac{2}{r} \left[ \left( 1 - \frac{3M}{r} \right) + \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \right] A_r - 2 \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r A_r \\ \Rightarrow & \quad \left( 1 - \frac{2M}{r} \right)^{-1} \partial_t^2 \Psi - \partial_r \left( \left( 1 - \frac{2M}{r} \right) \partial_r \Psi \right) + \frac{l(l+1)}{r^2} \Psi = -\frac{2}{r} \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \Psi \\ & + \frac{2}{r} \left[ \left( 1 - \frac{3M}{r} \right) + \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \right] A_r - 2 \frac{\frac{3\xi^2 M}{r^3} (1 - \frac{2M}{r})}{1 + \frac{2\xi^2 M}{r^3}} \partial_r A_r \end{aligned} \quad (40)$$

$$\begin{aligned}
& \left(1 - \frac{2M}{r}\right)^{-1} \partial_t^2 \frac{\left(1 - \frac{2M}{r}\right) \partial_r \Phi}{1 + \frac{2\xi^2 M}{r^3}} - \partial_r \left( \left(1 - \frac{2M}{r}\right) \partial_r \left( \frac{\left(1 - \frac{2M}{r}\right) \partial_r \Phi}{1 + \frac{2\xi^2 M}{r^3}} \right) \right) \\
& + \frac{l(l+1)}{r^2} \frac{\left(1 - \frac{2M}{r}\right) \partial_r \Phi}{1 + \frac{2\xi^2 M}{r^3}} = -\frac{2}{r} \frac{\frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \left( \frac{\left(1 - \frac{2M}{r}\right) \partial_r \Phi}{1 + \frac{2\xi^2 M}{r^3}} \right) \\
& + \frac{2}{r} \left[ \left(1 - \frac{3M}{r}\right) + \frac{\frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)}{1 + \frac{2\xi^2 M}{r^3}} \right] \frac{l(l+1)\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} - 2 \frac{\frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \left( \frac{l(l+1)\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \right) \\
& \Rightarrow \partial_t^2 \partial_r \Phi - \left(1 + \frac{2\xi^2 M}{r^3}\right) \partial_r \left( \left(1 - \frac{2M}{r}\right) \partial_r \left( \frac{\left(1 - \frac{2M}{r}\right) \partial_r \Phi}{1 + \frac{2\xi^2 M}{r^3}} \right) \right) \\
& + \frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \partial_r \Phi = -\frac{2}{r} \frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right) \partial_r \left( \frac{\left(1 - \frac{2M}{r}\right) \partial_r \Phi}{1 + \frac{2\xi^2 M}{r^3}} \right) \\
& + \frac{2}{r} \left[ \left(1 - \frac{3M}{r}\right) \left(1 + \frac{2\xi^2 M}{r^3}\right) + \frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right) \right] \frac{l(l+1)\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \\
& - 2 \frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right) \partial_r \left( \frac{l(l+1)\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \right) \\
& \Rightarrow \partial_t^2 \partial_r \Phi - \partial_r \left( \left(1 - \frac{2M}{r}\right) \partial_r \left( \left(1 - \frac{2M}{r}\right) \partial_r \Phi \right) \right) + \frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \partial_r \Phi \\
& - \frac{2}{r} \left(1 - \frac{3M}{r}\right) \frac{l(l+1)\Phi}{r^2} = \left(1 + \frac{2\xi^2 M}{r^3}\right) \partial_r \left( \left(1 - \frac{2M}{r}\right) \partial_r \left( \frac{\left(1 - \frac{2M}{r}\right) \partial_r \Phi}{1 + \frac{2\xi^2 M}{r^3}} \right) \right) \\
& - \partial_r \left( \left(1 - \frac{2M}{r}\right) \partial_r \left( \left(1 - \frac{2M}{r}\right) \partial_r \Phi \right) \right) - \frac{2}{r} \frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right) \partial_r \left( \frac{\left(1 - \frac{2M}{r}\right) \partial_r \Phi}{1 + \frac{2\xi^2 M}{r^3}} \right) \\
& - \frac{2}{r} \left(1 - \frac{3M}{r}\right) \frac{l(l+1)\Phi}{r^2} + \frac{2}{r} \left(1 - \frac{3M}{r}\right) \frac{l(l+1)\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \\
& + \frac{2}{r} \frac{\xi^2 M}{r^3} \left(5 - \frac{12M}{r}\right) \frac{l(l+1)\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} - 2 \frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right) \partial_r \left( \frac{l(l+1)\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \right) \quad (41)
\end{aligned}$$

$$\begin{aligned}
& \partial_t^2 \partial_r \Phi - \partial_r \left( \left(1 - \frac{2M}{r}\right) \partial_r \left( \left(1 - \frac{2M}{r}\right) \partial_r \Phi \right) \right) + l(l+1) \partial_r \left( \frac{1}{r^2} \left(1 - \frac{2M}{r}\right) \Phi \right) \\
&= \partial_r \left( \frac{2}{r} \frac{\frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)^2}{\left(1 + \frac{2\xi^2 M}{r^3}\right)} \partial_r \Phi \right) - l(l+1) \partial_r \left( \frac{2}{r^2} \frac{\frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)}{\left(1 - \frac{4\xi^2 M}{r^3}\right)} \Phi \right) \\
&\Rightarrow \partial_t^2 \Phi - \left(1 - \frac{2M}{r}\right) \partial_r \left( \left(1 - \frac{2M}{r}\right) \partial_r \Phi \right) + \frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \Phi \\
&= \frac{2}{r} \frac{\frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)^2}{\left(1 + \frac{2\xi^2 M}{r^3}\right)} \partial_r \Phi - \frac{2l(l+1)}{r^2} \frac{\frac{3\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)}{\left(1 - \frac{4\xi^2 M}{r^3}\right)} \Phi \\
&\Rightarrow \partial_t^2 \Phi - \left(1 - \frac{2M}{r}\right) \left(1 + \frac{2\xi^2 M}{r^3}\right) \partial_r \left( \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \Phi \right) = -\frac{l(l+1)}{r^2} \frac{1 + \frac{2\xi^2 M}{r^3}}{1 - \frac{4\xi^2 M}{r^3}} \left(1 - \frac{2M}{r}\right) \Phi
\end{aligned} \tag{42}$$

But we have to check and make sure this is consistent with the  $A_r$  equation:

$$\begin{aligned}
& \left(1 - \frac{2M}{r}\right)^{-1} \partial_t^2 A_r - \frac{1}{r^2} \partial_r \left( r^2 \left(1 - \frac{2M}{r}\right) \partial_r A_r \right) + \frac{l(l+1)}{r^2} A_r - \frac{2}{r^2} \partial_r \left[ r \left(1 - \frac{2M}{r}\right) A_r \right] \\
&+ \frac{1}{r^2} \partial_r \left( r^2 \frac{\frac{6\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)}{1 + \frac{2\xi^2 M}{r^3}} \partial_r A_r \right) = \frac{\frac{6\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)^{-1}}{1 + \frac{2\xi^2 M}{r^3}} \partial_t^2 A_r \\
&\Rightarrow \frac{l(l+1)}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \left(1 - \frac{2M}{r}\right)^{-1} \partial_t^2 \Phi - \frac{1}{r^2} \partial_r \left( r^2 \left(1 - \frac{2M}{r}\right) \partial_r \left( \frac{l(l+1)\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \right) \right) \\
&+ \frac{l(l+1)}{r^2} \frac{l(l+1)\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} - \frac{2}{r^2} \partial_r \left[ r \left(1 - \frac{2M}{r}\right) \frac{l(l+1)\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \right] \\
&+ \frac{1}{r^2} \partial_r \left( r^2 \frac{\frac{6\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \left( \frac{l(l+1)\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \right) \right) = \frac{\frac{6\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)^{-1}}{1 + \frac{2\xi^2 M}{r^3}} \frac{l(l+1)}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \partial_t^2 \Phi \\
&\Rightarrow \frac{\left(1 - \frac{2M}{r}\right)^{-1}}{r^2 \left(1 + \frac{2\xi^2 M}{r^3}\right)} \partial_t^2 \Phi - \frac{1}{r^2} \partial_r \left( r^2 \left(1 - \frac{2M}{r}\right) \partial_r \left( \frac{\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \right) \right) \\
&+ \frac{l(l+1)}{r^2} \frac{\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} - \frac{2}{r^2} \partial_r \left[ r \left(1 - \frac{2M}{r}\right) \frac{\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \right] \\
&+ \frac{1}{r^2} \partial_r \left( r^2 \frac{\frac{6\xi^2 M}{r^3} \left(1 - \frac{2M}{r}\right)}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \left( \frac{\Phi}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)} \right) \right) = 0 \\
&\Rightarrow \partial_t^2 \Phi - \left(1 - \frac{2M}{r}\right) \left(1 + \frac{2\xi^2 M}{r^3}\right) \partial_r \left( \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \Phi \right) = -\frac{l(l+1)}{r^2} \frac{1 + \frac{2\xi^2 M}{r^3}}{1 - \frac{4\xi^2 M}{r^3}} \left(1 - \frac{2M}{r}\right) \Phi
\end{aligned} \tag{43}$$

Yes! the equations are consistent

## 1.5 To Summarize

First Solution:

$$A_\mu = \{0, 0, \partial_z \Phi_1, -\partial_z \Phi_1\} \quad (44)$$

where

$$\partial_t^2 \Phi_1 - \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \left( \left(1 - \frac{2M}{r}\right) \left(1 + \frac{2\xi^2 M}{r^3}\right) \partial_r \Phi_1 \right) = -\frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \frac{1 - \frac{4\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \Phi_1 \quad (45)$$

The second solution:

$$A_\mu = \left\{ 0, \frac{l(l+1)\Phi_2}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)}, \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}} \partial_z \partial_r \Phi_2, \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}} \partial_z \partial_r \Phi_2 \right\} \quad (46)$$

where

$$\partial_t^2 \Phi_2 - \left(1 - \frac{2M}{r}\right) \left(1 + \frac{2\xi^2 M}{r^3}\right) \partial_r \left( \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \Phi_2 \right) = -\frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \frac{1 + \frac{2\xi^2 M}{r^3}}{1 - \frac{4\xi^2 M}{r^3}} \Phi_2 \quad (47)$$

## 1.6 WKB Approximation

Let  $\Phi$  be either  $\Phi_1$  and  $\Phi_2$

$$\Phi(r, t) = A(r) e^{i\phi(r) - i\omega t} Y_l^m(\theta, \varphi) \quad (48)$$

The general structure of the kinds of equations we want to solve is

$$\partial_t^2 \Phi - q(r) \frac{\partial}{\partial r} \left( p(r) \frac{\partial \Phi}{\partial r} \right) = -\frac{l(l+1)}{r^2} q(r) H(r) \Phi \quad (49)$$

In the case of  $\Phi_1$

$$q(r) = \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}}, \quad p(r) = \left(1 - \frac{2M}{r}\right) \left(1 + \frac{2\xi^2 M}{r^3}\right), \quad H(r) = 1 - \frac{4\xi^2 M}{r^3} \quad (50)$$

In the case of  $\Phi_2$

$$q(r) = \left(1 - \frac{2M}{r}\right) \left(1 + \frac{2\xi^2 M}{r^3}\right), \quad p(r) = \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}}, \quad H(r) = \frac{1}{1 - \frac{4\xi^2 M}{r^3}} \quad (51)$$

No matter which equation we are talking about, if we substitute the ansatz into the general equation what we get is

$$\begin{aligned} & -\omega^2 A e^{i\phi} - q \partial_r [p(A' + iA\phi') e^{i\phi}] = -\frac{l(l+1)}{r^2} q H A e^{i\phi} \\ \Rightarrow & \omega^2 A + q [p'(A' + iA\phi') + p(A'' + iA'\phi' + iA\phi'') + ip(A' + iA\phi')\phi'] = \frac{l(l+1)}{r^2} q H A \end{aligned} \quad (52)$$

Assuming that  $A$  and  $\phi$  are real functions

$$A'' - A\phi'^2 + \frac{p'}{p}A' = \frac{l(l+1)}{r^2} \frac{H}{p} A - \frac{\omega^2}{qp} A \quad (53)$$

$$A\phi'' + 2A'\phi' + \frac{p'}{p}A\phi' = 0 \quad (54)$$

$$\frac{1}{p} \frac{d}{dr}(pA') - A\phi'^2 = \left[ \frac{l(l+1)}{r^2} \frac{H}{p} - \frac{\omega^2}{qp} \right] A, \quad \frac{d}{dr}(pA^2\phi') = 0 \quad (55)$$

If  $\frac{d}{dr}(pA') \approx 0$  then

$$pA^2\phi' = C^2 \quad (56)$$

where  $C$  is a constant

$$\begin{aligned} -A \left( \frac{C^2}{pA^2} \right)^2 &= \left[ \frac{l(l+1)}{r^2} \frac{H}{p} - \frac{\omega^2}{qp} \right] A \Rightarrow \frac{C^4}{p} = \left[ \frac{\omega^2}{q} - \frac{l(l+1)}{r^2} H \right] A^4 \\ \Rightarrow A &= C \left( \frac{q/p}{\omega^2 - \frac{l(l+1)}{r^2} qH} \right)^{1/4} \end{aligned} \quad (57)$$

$$\phi' = \frac{C^2}{pA^2} = \frac{1}{p} \sqrt{\frac{p}{q} \left[ \omega^2 - \frac{l(l+1)}{r^2} qH \right]} = \frac{1}{\sqrt{qp}} \sqrt{\omega^2 - \frac{l(l+1)}{r^2} qH} \quad (58)$$

Putting this all together, for the  $\Phi = \Phi_1$  case

$$\begin{aligned} \Phi_1 &= C_{\omega lm}^{(1)} e^{-i\omega t} Y_l^m(\theta, \varphi) \left( \frac{q/p}{\omega^2 - \frac{l(l+1)}{r^2} qH} \right)^{1/4} \exp \left[ \pm i \int dr \frac{1}{\sqrt{qp}} \sqrt{\omega^2 - \frac{l(l+1)}{r^2} qH} \right] \\ &= \frac{C_{\omega lm}^{(1)} e^{-i\omega t} Y_l^m(\theta, \varphi)}{\sqrt{1 + \frac{2\xi^2 M}{r^3}}} \left( \omega^2 - \frac{l(l+1)}{r^2} \left( 1 - \frac{2M}{r} \right) \frac{1 - \frac{4\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \right)^{-1/4} \\ &\quad \times \exp \left[ \pm i \int dr \frac{1}{|1 - \frac{2M}{r}|} \sqrt{\omega^2 - \frac{l(l+1)}{r^2} \left( 1 - \frac{2M}{r} \right) \frac{1 - \frac{4\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}} \right] \end{aligned} \quad (59)$$

On the other hand, for the  $\Phi = \Phi_2$  case

$$\begin{aligned} \Phi_2 &= C_{\omega lm}^{(2)} e^{-i\omega t} Y_l^m(\theta, \varphi) \sqrt{1 + \frac{2\xi^2 M}{r^3}} \left( \omega^2 - \frac{l(l+1)}{r^2} \left( 1 - \frac{2M}{r} \right) \frac{1 + \frac{2\xi^2 M}{r^3}}{1 - \frac{4\xi^2 M}{r^3}} \right)^{-1/4} \\ &\quad \times \exp \left[ \pm i \int dr \frac{1}{|1 - \frac{2M}{r}|} \sqrt{\omega^2 - \frac{l(l+1)}{r^2} \left( 1 - \frac{2M}{r} \right) \frac{1 + \frac{2\xi^2 M}{r^3}}{1 - \frac{4\xi^2 M}{r^3}}} \right] \end{aligned} \quad (60)$$